

Learning Assisted Interactive 3D modelling from 3D sketches

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ABSTRACT

Sketching is one of the most natural ways for representing any object pictorially. With the advent of Virtual Reality (VR) and Augmented Reality (AR) technologies, 3D sketching has become more accessible. But it is challenging to convert these sketches to 3D models in a manner consistent with the intent of the artist. Learning based methods provide an effective alternate paradigm for solving this classical problem of sketch based modelling in interactive platforms. Surface patches, inferred from 3D sketch strokes, can be treated as a primitive which are assembled to create complex object models. We present our proposed framework for this problem and discuss our solution to one of its core challenges. Our deep neural network based method allows users to create surfaces from a stream of sparse 3D sketch strokes. We also show integration of our method into an existing Blender based 3D content creation pipeline. This serves as a basis to solve a more complex problem of creating complete 3D object models from sparse 3D sketch strokes.

CCS CONCEPTS

• **Computing methodologies** → **Parametric curve and surface models**; *Supervised learning*; *Virtual reality*; **Mesh models**.

KEYWORDS

3D sketches, deep neural networks, object modelling, parametric surfaces

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1 INTRODUCTION

Artists have leveraged sketches to model the world around them for decades. Recent advances in AR/VR technologies are allowing us to re-imagine the way we look at the classical problems in computer graphics like interactive 3D modelling. Interfaces based upon AR/VR strive to translate the intuitiveness of sketching using a pen-and-paper to 3D content creation. Compared to its 2D counterpart, 3D sketches have more expressive power as they eliminate the issues inherent in 2D to 3D translations. However, 3D sketches also come with their own issues of inconsistency present in the

sketch strokes due to higher degrees of freedom present in the 3D sketching environment. Thus, model creation from 3D sketches is a non-trivial endeavour.

A sketch based modelling framework, in general, takes sketches as input and can produce static 3D models or dynamic animations as output. For creating 3D models we must generate their geometry. Generating the geometry or shape of 3D models from given 3D strokes involves interpreting each of these strokes and building a compatible and feasible feature set that can be used to construct the model. Over the years, several approaches have been designed for constructing the 3D models. From our survey paper on sketch based content creation [5], we note that optimization based techniques have been widely explored. Learning based techniques offer an alternate paradigm for solving this due to their ability to scale and diversify to new problem settings. It is an open problem therefore, to parse the general intent of the artist from a sparse set of sketched strokes. The method should be amenable to providing a feedback and intuitive control to both novice and expert artists for ease of creation.

The problem that we solve in this thesis is to design an interactive learning based method that incorporates artist feedback to facilitate object-class agnostic modelling of 3D objects from sparse 3D sketches. The method must be easily amenable for integration into standard modelling pipelines.

2 BACKGROUND

Several approaches have been designed for constructing the 3D models using sketch input varying from annotated 2D sketched strokes to freehand 3D sketched strokes. Deering et al. [10] proposed one of the earliest VR based sketching system called HoloSketch. Users are allowed to create 3D shapes in the mid-air to edit and animate the models created. Many approaches like [11, 15, 19] are designed to minimize the user input by modeling from a single view sketch and creating corresponding models with their parts meaningfully positioned at appropriate depths automatically. In contrast to the above, methods like [2, 20] require the user to sketch strokes from multiple views and are able to create more complex models. Another category of approaches, for example [1] minimizes the required user input to only two strokes but significantly reduces

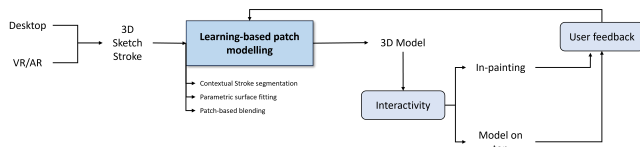


Figure 1: Our proposed 3D sketch based modelling framework.

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the set of objects that can be modelled from the strokes. Complementary to these methods, some techniques like [17, 21, 24, 25] can create a large variety of models at the cost of the user being forced to sketch a large number of strokes. Their technique is based upon progressively modelling with patches from curve networks and loops while preserving the geometric features. Alternately, many methods like [9, 22] propose modelling based on reference models where a base model is used to determine the positioning of strokes. Methods like [13, 14, 26] design learning based architectures to model some specific classes of objects. Their core idea lies in deriving the feature set specific to the target object class. Our survey [5] presents more details on various aspects of sketch based content creation including 3D modelling. As we can infer from these different categories of methods, the methods either are object class specific, or lack in processing an incoming stream of sparse 3D free-form strokes to create plausible 3D patch based models.

3 FRAMEWORK

The major requirements for a sketch based modelling framework is: (1) Process sparse set of 3D sketch strokes, (2) Object class-agnostic, (3) Interactivity, (4) Feedback based, (5) Easy integration into standard pipeline.

Sketches are generally sparse in nature as they are meant to be a minimal and quick representation of the objects. Processing these 3D strokes to produce plausible models requires an understanding of the shape that they represent. Also, estimating a shape from a set of strokes is an under-constrained problem with a wide range of mathematically valid but unrealistic solutions. This is due to the fact that different set of sketch strokes could correspond to roughly the same shape. Therefore, it is important to understand the inter-strokes semantics as well. A desirable feature of any modelling framework is generalize to a wide class of objects and ensure smooth interactivity. Both of these require a modelling construct which approximates a wide range of shapes along with feasibility of being modified easily and quickly. For this, a parametric representation of the input is handy in general. This brings us to the approach of primitive based modelling, which is quite common in the standard modelling pipeline. Primitives may include geometric constructs like cones, spheres, cylinders, cuboid, Bézier or BSpline surfaces and others. All of them are represented with few parameters which are modified to produce the required change in the representation. From the literature, we can say that parametric surfaces are one such geometric primitive which can approximate a wide range of shapes and has a well defined parameterization. Also, a collection of such surfaces is more efficient which can represent more complex objects as compared to a single surface. Feedback based systems are vital for making it beneficial to both novice and expert users. This gives control over the kind of output produced to the user, as well as helps in improving the output of the system. Particularly, it helps a lot in producing accurate output when there is a range of possible solutions for to a given input. We choose to have visual feedback based system as we target the 3D (AR/VR) platforms in general. Visual feedback ranges from providing visual assistance in drawing 3D sketch strokes to correcting the output surface(s) for improvement in future output. Lastly, to further enhance the usability of the framework, it should be compatible with standard

modelling pipelines like Blender [8]. This makes it easy for the users to leverage the advantages offered by the framework in an already known workflow.

We propose a framework (Figure 1) which includes all these aspects while using learning based techniques to solve the core challenges. From this proposed framework, we can see that fitting a surface patch to a given set of 3D strokes is central to it. We will highlight this particular aspect of the framework in the next section. It involves fitting Bézier surface from input 3D sketch strokes using a neural network architecture from our paper [6]. We also show a potential way of integrating our solution to the Blender pipeline for the task of 3D in-painting given a patch based 3D model. This task of in-painting is part of the framework as well as it is essentially a use case of the interactive modification of a given model.

4 PATCH BASED MODELLING FROM 3D SKETCH

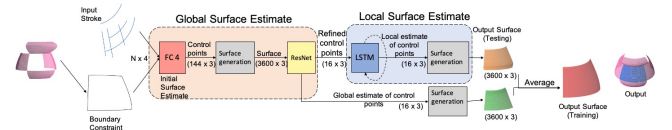


Figure 2: Overview of StpNet architecture: (1) *Global surface estimate* takes the 3D sketch strokes (with optional neighboring patch boundaries for blending) and produces its 3D surface approximation. (2) *Local surface estimate* refines the predicted 3D surface to incorporate the changes in shape with each incoming stroke. (3) An additional *post processing* step stitches the predicted 3D surface with the neighboring patches.

4.1 StpNet: Patch Fitting on 3D Strokes

We propose an end-to-end neural network model **StpNet** (Figure 2), which can fit a Bézier patch to an incoming stream of 3D sketch strokes. We represent the inferred 3D patches as bicubic Bézier surface. The input to the model is a sequence of strokes $S = \{s_k\}_{k=1}^N$ such that $s_k = \{p_i\}_{i=1}^n$ where $p_i = (x, y, z)$ is a point in 3D space.

We choose to predict the control grid corresponding to the output Bézier surface from our model and then compute the surface from it. The global surface estimator comprises of an initial surface estimator with a series of fully connected layers to predict a starting estimate of the patch corresponding to the strokes input till then. This is followed by ResNet based layers, which refines the initial surface such that it fits to all the strokes given till that point of time while trying to capture the overall shape that the strokes represent. Since the sketch strokes are added in an incremental manner, there is a local characteristic associated with the estimation as the sketch progresses. We pass the output of the global surface estimator to the local feature based surface estimator, which consists of LSTM layers. This takes into account the surface fitted to the previous strokes and predicts the current surface. The average of these two surfaces is taken as the final output from the network while training. The surface estimated finally at the end of the entire network is considered as output at test time.

In order to join a surface patch smoothly with adjoining patches, we need the common boundaries from the adjacent patches. For this, we maintain a global graph based data structure of all the patches from the object to find its four closest neighbors. We maintain a local octree based structure to determine the relative orientation of these neighbors to the input strokes. These boundaries along with the input strokes are given to our network to produce a patch which nearly blends to the rest of the object. To ensure smooth connectivity, we stitch the boundaries of the predicted patch to the corresponding ones from the adjacent patches.

4.1.1 Dataset. We use ABCSpline and boundABCSpline datasets of strokes and corresponding surface, which we derived from ABC dataset [12] as mentioned in our paper [6].

4.1.2 Preprocessing: We fit a cubic 3D Bézier curve to each stroke in the datapoint for smoothing out jitter in the input strokes. We randomly shuffle the order of the strokes to make the model stroke order agnostic.

4.1.3 Loss Functions. We use a modified one sided Chamfer distance [4] to measure the difference between the ground truth surface and the predicted surface. The L_2 norm is replaced with a variant of L_α -norm inspired from [3]. We use $\alpha = 1$ which represents a smoothed L_1 norm, and helps account for lack of information in regions where no input strokes are there.

$$\mathcal{L}_{\text{surf}} = \sum_{S \in D} \frac{1}{|S|} \sum_{x \in P} \|x - \hat{x}\|_\alpha \quad (1)$$

Here, S denotes a surface in dataset D and $\hat{x}$ is a point sampled from it, and x is a point sampled from the predicted surface P such that they correspond to the same grid point in UV space. $\|\cdot\|_\alpha$ denotes the α norm between two 3D points and $|\cdot|$ denotes the size of a set.

4.2 Results

We use both sided Chamfer distance [23] as the error metric for evaluation.

$$d(S, \hat{S}) = \max\left\{\frac{1}{|S|} \sum_{x \in S} \min_{\hat{x} \in \hat{S}} \|x - \hat{x}\|_2, \frac{1}{|\hat{S}|} \sum_{\hat{x} \in \hat{S}} \min_{x \in S} \|\hat{x} - x\|_2\right\} \quad (2)$$

We compute this distance for 3.6K points sampled from the predicted surface $\hat{S}$ and ground truth surface S , and report the average values in our results.

4.2.1 Patch Fitting Evaluation. We use the methods given in [18] (PsN), [23] (FdN) and cubic spline surface interpolation (Spline Optim (SOp)) [7] as baselines. We report the quantitative results in Table 1 and qualitative comparisons in Figure 3.

4.2.2 Interactive Surface Creation. We use the models from the Utah tea set [16] and randomly remove patches from these models. We predict these missing patches from the models (See Figure 5). We implement a Blender plug-in to integrate our method for in-painting models. We show examples of adding new patches with different stroke input in Figure 4.

4.2.3 Analysis. Our patch fitting method outperform various existing neural and non-neural methods for scattered data interpolation, showing its efficacy in solving the core problem. The results show that our proposed method integrates well with the existing 3D

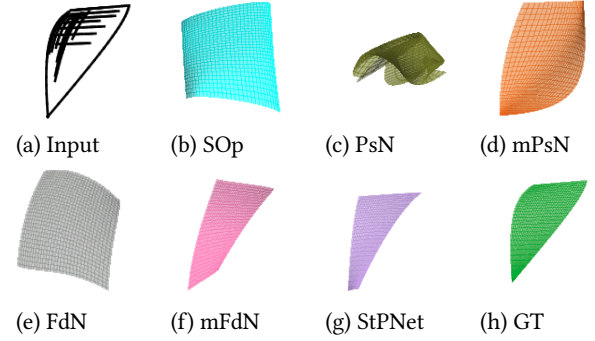


Figure 3: Qualitative results for patch fitting on boundABCSpline dataset. (a) input strokes (b)-(g) represent predicted patch from the mentioned methods. (b) cubic interpolation [7] (c) parsenet [18] (d) modParsenet (e) foldingnet [23] (f) modFoldingnet (g) StPNet (our method) (h) Ground truth patch.

Table 1: Comparison between various methods and our method on ABCSpline and boundABCSpline datasets for a Bézier surface patch fitting. Loss: $\alpha=2$:original Chamfer loss, $\alpha=1$:modified Chamfer loss. The percentage improvement of our method over the best performing baselines: ABCSpline: 31.91%, boundABCSpline: 91.46%.

Method	Boundary strokes	Initial surface	Loss α	Error
Spline Optim	✗	-	-	0.2639
Parsenet	✗	✗	2	0.1510
modParsenet	✗	✓	1	0.0683
Foldingnet	✗	✗	2	0.0141
modFoldingnet	✗	✓	1	0.0495
StPNet	✗	✓	1	0.0096
Spline Optim	✓	-	-	0.1317
Parsenet	✓	✗	2	0.1235
modParsenet	✓	✓	1	0.0306
Foldingnet	✓	✗	2	0.0051
modFoldingnet	✓	✓	1	0.0041
StPNet	✓	✓	1	0.00035

modelling pipeline in Blender for the use-case of in-painting patch based 3D models. The stability of our method with a sparse set of strokes can be handy for quick repair of a 3D model.

5 CONCLUSION AND FUTURE WORK

The discussed method shows the potential of learning based methods for modelling from 3D sketches. The input 3D strokes can be input from the desktop or an AR/VR platform. Our method for fitting surface patches to a sparse stream of 3D strokes paves the way for modelling of more complex objects. To this end, we are working on extending this patch based model creation to a more generic use case like modelling from scratch or on top on existing models. For that, we add a mechanism to learn importance of each stroke in

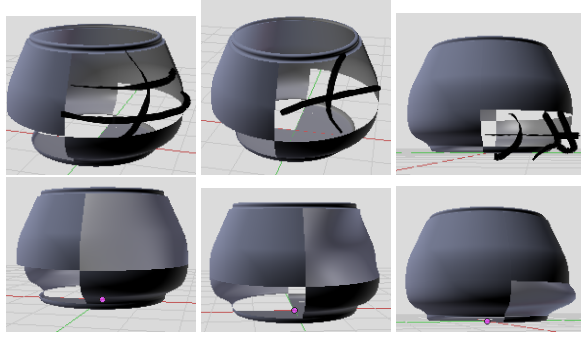


Figure 4: Results of in-painting in real time. Top row: Input stroke on incomplete object, Bottom row: Output surface with the object

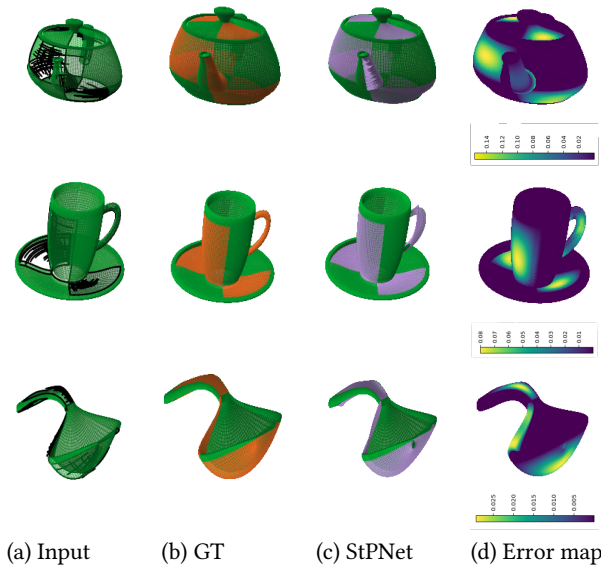


Figure 5: Qualitative results for editing patch fitting on Utah teapot models. (a) Incomplete model with input strokes (b) Patches in orange are ground truth for missing patches (c) Patches in lavender are predicted patches for the missing patches (d) Point-wise error between predicted and ground truth missing patches.

the final output on top of the existing architecture. This makes the system more robust to noise as well as gives appropriate weight to other information coming from corresponding neighborhood. This enables the method to create models for entire objects. We intend to make the framework an end-to-end solution by adding supporting functionality like converting different 3D models to patch based models, contextually segmenting 3D sketches into group of strokes, smoothly blending 3D surfaces together, and other similar functions. We hope that this framework will help artists in leveraging the benefits of 3D sketching by making the process more intuitive and easy to use.

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